

Ocean floor

The ocean floors are not just flat, featureless plains. The deep blue water of the oceans conceals a hidden landscape of shallow coastal seabeds, rocky reefs, vast muddy plains, incredibly deep chasms, and colossal volcanoes. Long ridges of high mountains stretch for many thousands of miles across the ocean floors, forming the longest mountain ranges on the planet. Until very recently, we had no idea that many of these features existed, or why they were there.

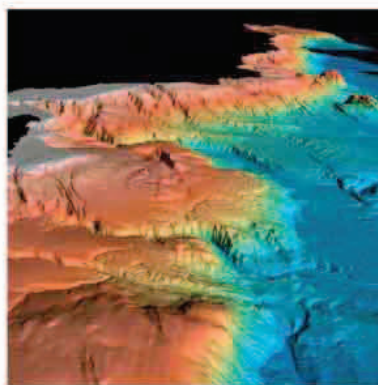
WOW!

There are ocean trenches that are deep enough to swallow some of the tallest mountains on Earth.

UNDERWATER WORLD

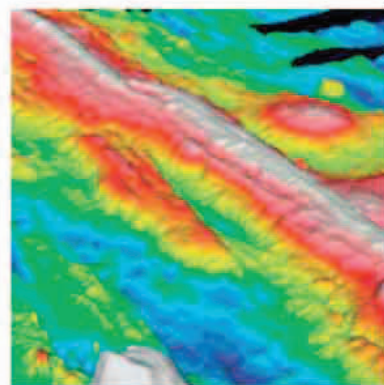
As methods of measuring depth have improved, scientists have been able to detect more features of the ocean floor. This section through a typical ocean shows the most important features, along with images that were gathered using the most advanced technology. Color-coded for depth, they reveal a hidden world beneath the waves.

Continent, fringed by the shallow continental shelf



▲ CONTINENTAL SHELF

The shallow regions at the fringes of oceans are the continental shelves. At the edge of the shelf, the continental slope descends to the deep ocean floor. This image shows the shallow shelf in red, and the ocean floor in blue.



▲ OCEAN RIDGE

Made using echo-sounding technology, this image shows part of the long ridges (red) that snake across the ocean floors. Forming a network that extends around the globe, these ridges can be up to 3,300 ft (1,000 m) high.

Submerged seamount

